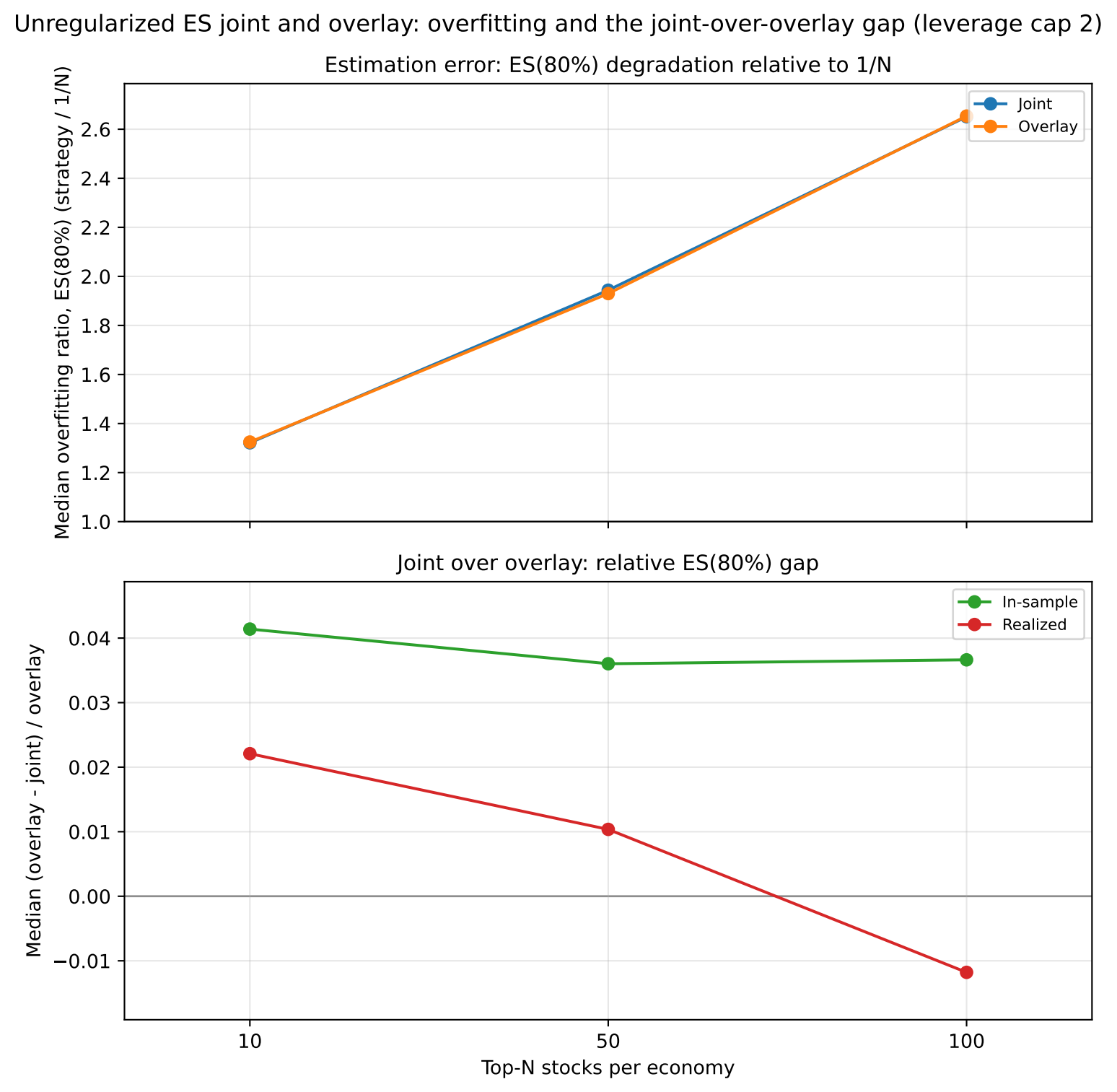
Unregularized ES joint and overlay: overfitting and the joint-over-overlay gap (leverage cap 2)
Estimation error: ES(80%) degradation relative to 1/N
Median overfitting ratio, ES(80%) (strategy / 1/N)
Joint
Overlay
1.0
1.2
1.4
1.6
1.8
2.0
2.2
2.4
2.6
Joint over overlay: relative ES(80%) gap
Median (overlay - joint) / overlay
In-sample
Realized
0.04
0.03
0.02
0.01
0.00
−0.01
10
50
100
Top-N stocks per economy